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ORIGINAL ARTICLE

Characterizations of laser transmission welding of glass and copper using nanosecond pulsed laser

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Abstract
In this study, we conducted an investigation into the heterogeneous bonding process of glass and copper using a nanosecond pulsed laser for laser transmission welding. Our research focused on examining various processing parameters and the influence of focal plane positions on bonding quality. To evaluate weld strength, we employed both pull-tensile separation force (PTSF) and shear-tensile separation force (STSF) measurements. The analysis of fracture and separation results encompassed detailed examinations of weld morphology, microstructure, elemental composition, and phase configuration. The results revealed that increasing the laser welding energy initially enhanced the weld strength until it reached a saturation point. This saturation point was attributed to the formation of voids or cracks, leading to residual stress within the weld zone due to non-uniform hot spots in the molten area during processing. Among the different focal plane positions tested, positioning it below the glass/copper interface yielded the highest weld strength. The maximum achieved bond strength was above 10 MPa, demonstrating the feasibility of cost-effective pulsed laser welding for copper-to-glass applications. Moreover, the weld strength obtained using STSF surpassed that of PTSF. This discrepancy arises from the fact that PTSF separation led to fractures at the weld seam-glass boundary, leaving the weld seam on the copper plate. In contrast, STSF resulted in separation along the weld-copper interface, leaving the weld seam on the glass sheet. In-depth analysis through SEM and EDS elucidated that Cu and SiO_2 underwent intra-mixing and inter-particle diffusion in the molten pool during welding. HR-TEM and SAED observations unveiled the presence of polycrystalline copper nanoparticles, copper oxides, and an amorphous Cu–O-Si region at the weld interface. This amorphous region significantly contributed to the robust bonding between copper and glass.

Keywords Nanosecond pulsed laser · Laser transmission welding · Heterogeneous junction · Glass-copper bonding · Bonding and separation mechanisms

1 Introduction

Glass has always been a highly significant material due to its numerous favorable application properties. These include low dielectric constant, high electrical resistivity, high corrosion resistance, high stiffness, and a low coefficient of thermal expansion. Consequently, glass has continuously served as an unlimited and innovative material, playing an essential role in the development of numerous products including microelectromechanical systems (MEMS) [1], photonics [2], microfluidics [3], optoelectronics [4], and optical telecommunications [5]. On the other hand, copper, with its excellent strength, ductility, electrical conductivity, and thermal conductivity, is another crucial metal with a longstanding history of usage. These characteristics have contributed to its extensive application across diverse industries and fields.

More recently, the need for joining of the two heterogeneous materials, glass and metal, has emerged in various fields, including microfluidic devices, sensors, MEMS devices, and microelectronic devices [6–8]. However, achieving precise and robust joining of glass and metal has always posed challenges due to the significant disparities in their material properties. Glass exhibits brittleness, while metal demonstrates ductility. Moreover, glass possesses high heat insulation characteristics, whereas metal exhibits high thermal conductivity. These material property differences make

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it difficult to find a practical, reliable, and cost-effective technique for precise and robust bonding between these two materials.

Various methods have been utilized for bonding glass and metal, including adhesive bonding, mechanical fixation, high-temperature sintering, and laser welding [9, 10]. Laser welding has emerged as a notable technique, garnering considerable attention due to its unique capability to deliver precise and localized heat for pattern bonding purposes. One of the advantages of laser welding is its ability to facilitate internal, pattern welding through a technique called laser transmission welding. This technique involves the laser beam passing through the upper glass and interacting with the metal in close contact with the glass, enabling effective and efficient bonding. Furthermore, laser welding offers the advantage of easy automation and digitalization [11, 12]. This characteristic allows for seamless integration into automated manufacturing processes, enabling precise control and monitoring of the bonding parameters. The digitalization of laser bonding further facilitates process optimization, quality assurance, and reproducibility, making it an attractive choice for industrial applications.

Ultrashort pulse lasers, such as femtosecond (fs) laser [6, 8, 13–15] and picosecond (ps) laser [16, 17] have recently garnered significant focus for bonding applications due to their exceptional precision in delivering welding energy within a small and well-defined range. These lasers generate a minimal heat-affected zone, making them suitable for precise and effective bonding. For example, Zhang et al. [8] explored the relationship between processing energy, scanning speed, and shear strength in fs laser-welded Cu/SiO_2 samples, achieving a maximum bonding strength of 2.34 MPa. Wang et al. [15] employed a fs laser to weld stainless steel to glass, achieving an elevated shear strength of approximately 8.79 MPa. Furthermore, Li et al. [13] delved into the welding of copper and glass using ultrashort pulse lasers and attained a peak welding strength of 10.9 MPa. These studies demonstrated the advancements made in laser bonding using ultrashort pulsed lasers and their potential for achieving substantial shear strength in welds.

One of the distinct merits of using ultra-short pulsed lasers in welding involving a transparent material is that they provide a unique energy source for precise regulation of the molten zone, particularly in scenarios where non-linear absorption effects are significant. However, several challenges still need to be addressed in their usage. These challenges include ensuring that the gap between the glass and metal is a few micro-meters range [14], their operating environment is very demanding [18, 19], and dealing with the complexity of the systems, which can lead to high costs. Moreover, due to the extremely high intrinsic peak power, the occurrence of defects in the resulting weld zone can be challenging to avoid, significantly impacting the welding quality and limiting its applicability. Consequently, it is crucial to carefully evaluate the appropriate bonding method for metal-to-glass applications.

A cost-effective alternative approach is to utilize longer pulse lasers, such as milisecond or nanosecond lasers while still retaining the advantages of laser transmission welding. For instance, Min et al. [20] investigated the bonding of glass and aluminum using a mini-second laser and examined the effects of laser power, scanning speed, and pulse repetition frequency on weld quality. They achieved a high shear strength of bonding, measuring 8.94 MPa. Li et al. [11] successfully conducted laser transmission welding of high borosilicate glass with titanium alloy using an Nd:YAG laser with a pulse width of 2.5 ms. Utsumi et al. [7] employed a nanosecond (ns) laser system and successfully weld borosilicate glass and copper. They obtained the maximum shear strength of the bond was 690 kPa according to a pulse energy of 300 μJ. In a recent study, Huo et al. [18] demonstrated that a strong weld between silica glass and 304 stainless steel, reaching up to 16.39 MPa, was achievable using an infrared ns laser. These studies highlight the feasibility of bonding glass and metal using longer pulse lasers, as long as appropriate laser processing parameters are carefully maintained.

The weld seam's shape and the location of fracture initiation are critical factors that impact the strength of a weld and require careful consideration. In the case of metal-to-glass welding with ultrashort pulsed lasers, the molten zone of the weld seam typically exhibits a peak-shaped, influenced by the extent and evolution of the molten zone during the welding process [19]. Regarding the location of fracture initiation, both studies from Tamaki et al. [21] and Zhang et al. [22] revealed that separation occurred along the boundary between the weld and the bottom glass when an external separation force was applied. Following separation, the weld was found to remain on the top glass.

In this study, we aim to analyze cross-sectional surface images obtained by transverse cutting of the weld seam to examine the shape of the resulting weld seams formed under various processing parameters. Additionally, we will investigate the fracture locations when subjected to two types of separation forces: pull-tensile separation force (PTSF) and shear-tensile separation force (STSF). The study will address the fracture mechanisms associated with these separation forces.

The quality of bonding is closely linked to the morphology, microstructure, elemental composition, and phase configuration around the interfaces of the weld and the matrix, as well as inside the weld. Previous studies have explored these aspects. For instance, Wang et al. [15] analyzed the surface morphology of the soda-lime glass/304 stainless steel interface using XRD analysis to predict potential mixing phenomena in the molten pool during the welding procedure. Cuica et al. [12] studied the mechanisms of weld formation and chemical reactions during the bonding of fused silica glass to aluminum using a ps laser through TEM-EDS analysis. Huo et al. [18] investigated the quality of ns laser bonding between silica glass and 304 stainless steel based on SEM and EDS mapping. They discovered the formation of the Fe–O-Si system, which they concluded greatly contributes to the improvement in joint strength.

In addition to addressing the aforementioned fracture mechanisms, this study also aims to understand the bonding performance of laser-bonded glass/copper welds using a cost-efficient ns laser system. Consequently, we will explore the relationships between weld quality and its formation processing parameters, such as laser power, scanning speed, and pulse repetition rate. This will be achieved through the examination and comparison of the geometries and morphologies within and around the periphery of the weld seam. SEM, EDS, and HR-TEM images will be utilized to characterize the weld's microstructures, elemental compositions, and phase configurations, enabling us to correlate them with the resulting bonding performance.

2 Experimental procedure

Referring to the flowchart presented in Fig. 1, an investigation was conducted to explore the welding of copper to glass using a ns laser. The study comprised three main stages: material preparation, laser welding, and characterization. In the initial stage, thorough cleaning of both the glass and copper surfaces designated for welding was performed to eliminate any potential contaminants. Subsequently, the samples were securely clamped and exposed to laser irradiation, facilitating the formation of a bond between the copper and glass. This stage constituted the process of laser welding. The third stage involved evaluating the bonding strength through a series of tests, including pull-tensile separation strength (PTSS) and shear-tensile separation strength (STSS) measurements. Additionally, the morphologies of the cross-sectional profile, perpendicular to the weld seam, and the tracks resulting from separation on both the glass substrate and copper plate surfaces were examined. After completing these assessments, the sample demonstrating the most satisfactory weld quality was selected for further microstructural investigation. This investigation encompassed the acquisition of SEM (Scanning Electron Microscopy), EDS (Energy-Dispersive X-ray Spectroscopy), HR-TEM (High-Resolution

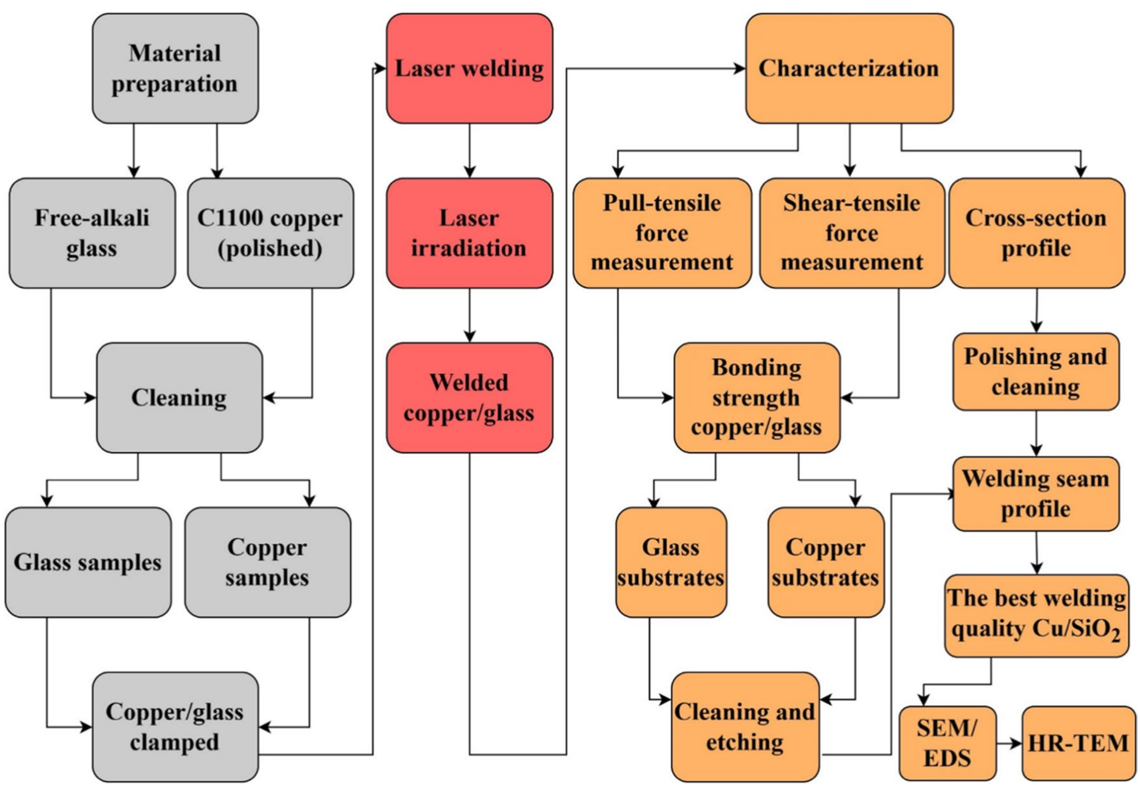

Fig. 1 Flowchart depicting the experimental process employed in this study

Transmission Electron Microscopy), and Selected Area Electron Scanning (SAED) images.

The subsequent sections provide detailed elaboration on each of the three stages.

2.1 Material preparation

2.1.1 Materials

A commercially available alkali-free glass and C1100 copper sheet were used for study. Their chemical compositions are listed in Tables 1 and 2, respectively. The glass sample for experiments was with a thickness of 0.7 mm and was cut into a rectangular shape, with a dimension of 75 mm in length and 25 mm in width. The experimental copper workpieces were prepared with the same dimensions in length and width as that of the glass, but their thickness was 3 mm. The thermal process in laser welding is significantly influenced by various material properties such as melting temperature, density, specific heat capacity, thermal expansion, and thermal conductivity. Table 3 shows physical properties of the as-purchased alkali-free glass and C1100 copper plates employed in this study.

2.1.2 Surface clearing

Precise welding generally requires low surface roughness and high surface cleanness. The ns laser welding, nevertheless, takes this to a much higher level. As a surface-surface direct joining method, it demands the gap between the surfaces to be on the micron or sub-micron scale [14]. Polished surfaces are thus typically required to generate a suitable small gap, preferably optical contact. The alkali-free glass surface used in this study is with a roughness of approximately 0.1 μm. These surfaces only require a deep cleaning step, to be described in the next step, for subsequent laser transmission bonding. The copper plate was processed by grinding its surface to a specified range of surface roughness, with R_a ranging from 0.1 to 0.3 μm. The surface roughness was obtained using a Confocal Laser Scanning Microscopy (CLSM), Keyence VK-X1000.

Before laser welding, it was necessary to remove any contaminants present on the surfaces of both the glass and copper. For the glass, a meticulous cleaning procedure was followed. It involved careful washing in acetone, followed by wiping with lint-free lens tissue. Subsequently, the glass was subjected to ultrasonic cleaning in isopropanol for 15 min. Finally, it was rinsed with deionized water and dried using nitrogen.

The copper samples underwent thorough cleaning using ethyl alcohol (C_2H_5OH) as a vapor phase-reducing agent. A combined cleaning process involving an ultrasonic bath was employed for 10 min to effectively clean accessible surface areas. Deionized water was used for washing, followed by drying to complete the cleaning process.

These cleaning procedures were implemented to remove contaminants from both the glass and copper surfaces, ensuring their readiness for the subsequent laser welding process.

2.2 Laser welding

2.2.1 Sample clamp

Maintaining a small gap between the two surfaces intended for welding is crucial to achieving successful results.

Table 1 Chemical composition of the alkali-free glass (wt.%)

SiO_2	Al_2O_3	B_2O_3	MgO	CaO	SrO	BaO
68–80	0–12	0–7	0–12	0–15	0–4	0–1

Table 2 Chemical composition of C1100 copper (wt.%)

Cu	O	Pb	S	Bi	Sb	As
≥99.9	≤0.06	≤0.005	≤0.005	≤0.002	≤0.002	≤0.002

Table 3 Physical properties of C1100 copper and alkali-free glass

Material properties	C1100 copper	Alkali-free glass
Coefficient of thermal expansion (1/K)	16.9×10^{-6}	3.2×10^{-6}
Thermal conductivity ((W/(m.K))	391.1 (20 °C)	1.09 (23 °C)
Melting point (°C)	1083	669 (Strain Point) 772 (Annealing point) 971 (Softening Point) 1293 (Working point)
Density (g/cm^3)	8.96 (20 °C)	2.38

However, during the welding process, the specific volume changes associated with the phase transition states of both glass and copper may cause an increase in the gap between the target surfaces. Thus, it is essential to employ an appropriate sample clamping method to preserve the desired small gap for the joining procedure. Various clamping methods for laser transmission welding have been explored in previous studies [6, 16]. Nevertheless, due to the demanding requirements for the fixture to withstand high pressure and accommodate the significant differences in material properties between glass and metal, the design and fabrication of suitable fixtures remain challenging.

In this particular research, a clamp structure was designed utilizing a load cell in conjunction with two stepper motors for precise control of the applied force. Figure 2 (a_1) depicts the photo of the clamp arrangement employed for the bonding process in this study. The system consisted of three ball-screw frames. One frame was driven by the stepper motors to secure them, while the second frame could be adjusted either forward or downward through the action of two screws. The final frame was utilized to position the load cell. The maximum achievable clamping force for this setup was 70 N. Experimental trials indicated that setting the clamping force below 50 N was inadequate to provide the necessary assistance during welding. Consequently, a consistent clamping force of 65 N was employed in the experiments.

2.2.2 Experimental setup

Referring to Fig. 2 (a), all laser welding experiments were conducted by a pulsed fiber laser (YLP-1–100-20–20, IPG)

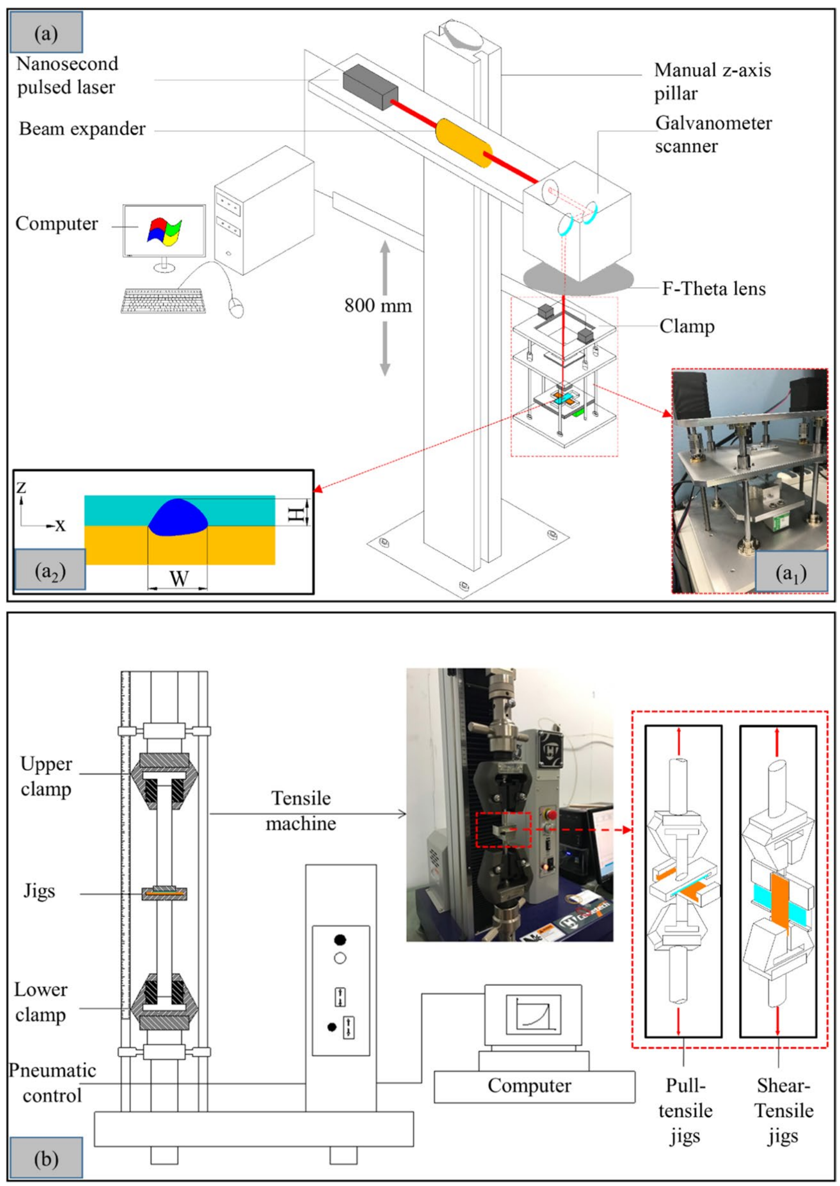

Fig. 2 Experimental setup for laser transmission bonding and weld strength measurement. **a** Schematic illustration of the ns laser processing system and the sample clamping system. Insets: (a_1) Photograph of the designed clamping system; (a_2) Cross-sectional view of a weld line, highlighting the width (W) and height (H) geometric terms and referential coordinates. **b** Tensile tester and associated jigs used to measure pull-tensile separation strength (PTSS) and shear-tensile separation strength (STSS)

that delivers near-infrared light at the wavelength of 1070 nm and with a pulse duration of 100 ns. Its maximum average power is 20 W. The welding system also includes mirrors for laser beam delivery, a focusing lens, and an X–Y worktable. The focal distance of the focusing lens was 292 mm and, after passing the lens, the laser beam was focused on a spot with a diameter of 40 µm on its focal plane. Notably, the plane of focus could be adjusted vertically using the manual Z-Axis 800 mm pillar attached to the Galvanometer scanner. This allowed for precise control of the laser spot's position along the Z-axis. Furthermore, a coaxial CCD camera was utilized to monitor the position of the laser spot on the focal plane.

Table 4 presents the five sets of processing parameters employed in the experiments. It is important to note that, generally, during laser processing, the three inputs, average laser power (P), scanning speed (v), and repetition rate (f), were not independent, but interrelated. Another factor employed to describe their interactions is pulse energy, represented as E, which is calculated as $E = \frac{P}{f}$ [23].

2.3 Characterizations

2.3.1 Weld line and weld seam properties

To facilitate a systematic comparison of the weld properties obtained using different laser processing parameters, the welding procedure was conducted as a line weld with a specified length of 17 mm. To mitigate potential end effects resulting from the starting and ending points of laser irradiation during the weld manufacturing process, all the characterized weld properties were obtained from the central portion of the line length. This approach ensures that the assessed properties are representative of the main welding region, minimizing any potential variations or distortions caused by the welding initiation and termination points.

The characterization of the weld area and surface topography involves examining the image of the cross-sectional surface, which is perpendicular to the line weld. Specifically, the X–Z plane in Fig. 2 (a_2) provides the view for analysis, enabling the assessment of the weld's extent and offering insights into various aspects of the surface features and morphology of the welded joint. By analyzing the cross-sectional image, valuable information regarding the structural integrity, bonding quality, and overall appearance of the weld can be obtained. For the transverse cut of the copper/glass weld seam, we utilized an IsoMetTM 4000 Linear Precision Saw and EXTEC Diamond Wafering Blades. To ensure optimal cutting conditions and minimize damage or distortions to the sample, distilled water with Buehler Cool 3 Cutting Fluid (33.8 oz) was used for cooling during the cutting process.

Table 4 Laser processing parameters

Experiment level	1	2	3	4	5
Average power (W)	12	13	14	15	16
Scanning speed (mm/s)	5	10	15	20	25
Frequency (kHz)	20	21	22	23	24

To enhance the image quality, we employed grinding and polishing techniques using the Buehler Ecomet 30 Semi-Automatic Grinder Polisher. The samples were sequentially ground using abrasive grinding papers with grit sizes of P1200, P2000, and P4000, at a rotational speed of 150 rpm. Subsequently, the surfaces were polished at an average speed of 50 rpm. Afterward, the polished samples were thoroughly cleaned using ethyl alcohol (C_2H_5OH) and subjected to a 10-min ultrasonic wash to remove any dirt or contaminants. Following the cleaning process, the cross-sectional surface images were captured using the CLSM technique.

Based on the images obtained from the CLSM, the dimensions of the weld seam were determined by referring to Fig. 2 (a_2) and measuring the width, W, and height, H. The original optical pictures of the welding seam were observed at a higher magnification of 50x. The positions in the interface area relevant to the weld size, W and H, were selected and marked. The mean values of the seam width and seam height were obtained by analyzing 5 cross-sectional areas of the weld. This approach ensured precise assessment and characterization of the weld seam's geometry, providing valuable dimensional information for further analysis.

To assess the geometry of the weld seam on both the glass and copper sides after separation (the methods for separating the weld seam will be explained in Sec. 2.3.2), the separated glass substrate and copper plate were immersed in a 10% Nital solution. The Nital solution was prepared by combining alcohol and nitric acid in a ratio of 10 ml of extra pure reagent-grade nitric acid (HNO_3) with 90 ml of ethyl alcohol (C_2H_5OH 99.5%, Nihon Shiyaku Reagent). The glass substrate and copper plate were immersed in the Nital solution for 2 min. This process allowed for the clear identification of the boundary between the weld seam and the glass/copper interface. The geometric and morphological features of the weld seam were then measured using the CLSM, providing valuable information about its structure and appearance.

2.3.2 Breaking strength

The bonding strength of the weld was assessed by measuring the force required to separate the glass substrate from the copper plate. Although several methods were available for this purpose, the majority focused on measuring tensile-separation strength [8, 24]. Referring to Fig. 2 (a_2), the PTSF was applied along the Z-axis, perpendicular to the weld seam line, while the STSF was applied along the Y-axis, parallel to the weld seam line. The tensile testing machine used was the QC-548M2F-Computerized Tensile/Compression Testing from Cometech Testing Machines Co., capable of applying a maximum tensile force of 5 kN and with a feeding speed ranging from 0.0002 to 3 mm/min. During the sample separation process, a digital dynamometer recorded the load as a function of position with an accuracy of 0.001%.

To ensure accurate measurements unaffected by factors such as vibration and eccentricity of the test piece during the separation process, it was crucial to design a specialized clamp suitable for the shape and size of the test piece and compatible with the testing equipment. The design of the clamping jig in the literature [25] served as a reference for designing the specific clamping jigs used in this study. The resulting clamping jigs for both the PTSF and STSF separation are schematically illustrated in the insets of Fig. 2 (b).

2.3.3 Weld morphology

The geometry, surface morphology, and cross-sectional profile of the bonded and separated weld seams were examined by the CLSM and a scanning electron microscope, SEM (JOEL, JSM-7000F). The elemental compositions around the copper/glass surface were examined using X-ray Diffraction (XRD), and the distribution maps were investigated using EDS BURKER XFlash®Detector 6060 at a voltage of 126 eV. A dual-beam focused ion beam, FIB (FEI, Versa 3D), was used to slice samples for the cross-sectional observation and analysis of the microstructures around the copper/glass junction area. The employed high-resolution transmittance electron microscope (HR-TEM) in this study was JEOL, JEM-2100.

3 Results and discussion

3.1 Bonding strength

The bonding strength plays a critical role in determining the quality of welding. Laser processing energy is one of the key processing parameters that significantly influence welding quality. Figure 3 illustrates the pull-tensile separation strength (PTSS) and shear-tensile separation strength (STSS) as a function of pulse energy for three different focal plane positions. The pulse energy was varied by adjusting the laser output power and pulse repetition rate, ranging from 0.5 to 0.8 mJ per pulse. The scanning speed was set at 5 mm/s, and the pulse frequency varied from 20 to 24 kHz, resulting in an average laser power of 12 to 16 W. The three focal plane positions were negative (0.5 mm below the interface), at the interface, and positive (0.5 mm above the interface), respectively. During processing, a fixed sample jig clamping force of 65 N was applied to provide pressure assistance.

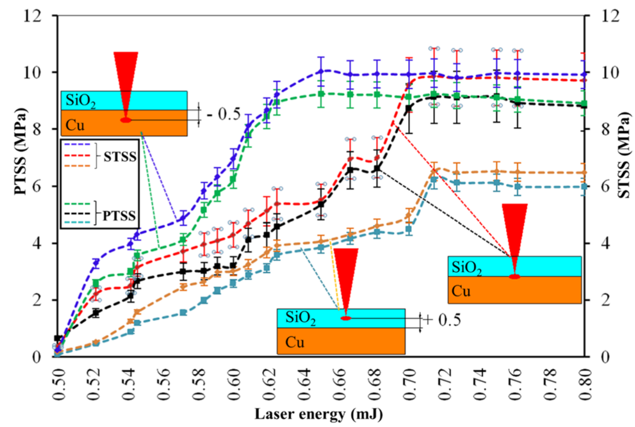

Fig. 3 Experimental results demonstrating the variation of pull-tensile separation strength (PTSS) and shear-tensile separation strength (STSS) with incident laser pulse energy

Both separation strengths exhibited a similar trend with increasing pulse energy. Initially, the strengths increased significantly, reached a maximum, and then tended to saturate as the pulse energy continued to increase. The results indicate that the STSS was generally higher than the PTSS for the sample weld under the same processing conditions and focal plane position. This difference can be attributed to the fracture or separation locations caused by the two separation methods. The pull-tensile separation resulted in a fracture occurring at the boundary between the weld and the glass, while the shear-tensile separation resulted in a fracture at the boundary between the weld and the copper. Additionally, a notable characteristic of ns laser welding was that the weld zone was predominantly located on the glass side, leading to a significantly larger weld/glass interface area compared to the weld/copper interface. This disparity in interface areas also contributed to the magnitudes of STSS and PTSS. Further discussion on this aspect will be provided in Sect. 3.2.

To enhance welding efficiency, it is crucial to effectively utilize laser energy. The focal plane position plays a critical role in distributing laser energy between the upper and lower components being welded, namely glass and copper, respectively. This distribution directly impacts the formation of the weld pool and influences the size, shape, and microstructure of the weld zone. Consequently, several researchers have investigated the influence of focal plane position on laser welding quality for different materials using various types of lasers. In Fig. 3, a comparison of the three focal plane positions showed that the negative position exhibited the highest welded strength. Previous studies on glass/glass welding using fs lasers have suggested that the favorable focal plane position is usually below the interface to obtain higher bonding strength [26, 27]. For bonding glass with non-transparent materials using a ps laser or fs laser, the optimal focal plane position could be either above, at, or below the interface, depending on other process parameters [16, 28]

The welding process involves utilizing laser energy to generate a molten pool at the interface of the two substrates being joined. Subsequently, the molten zone solidifies as it cools after the laser pulses pass. In the case of glass/glass welding, where glass is highly transparent to laser light, the focal plane position needs to be positioned near the glass/glass interface to enable nonlinear absorption induced by ps or fs lasers. However, this type of welding is not suitable for a ns laser. In this study, laser transmission welding was employed, where the majority of the incident laser energy was absorbed around the laser-irradiated area of the copper substrate. The absorbed energy was then transferred to the surrounding regions through heat conduction. The targeted region experienced continuous heating due to the repetitive laser pulses. When the local temperature reached a sufficiently high level, both the glass and copper would melt, facilitating their mixing within the molten zone through diffusion and convection mechanisms. After the laser irradiation ceased, the molten material solidified to form a weld zone. The resulting bond strength is influenced by the quality, size, morphology, composition, and microstructure of the weld zone.

While glass demonstrates high transparency to incident laser irradiation, it retains a portion of the transmitted laser energy. Additionally, its nonlinear absorption effect intensifies with rising temperatures. Placing the focal plane below the interface served to alleviate glass absorption, enabling the copper to absorb the majority of the laser energy and promoting a more pronounced temperature increase, facilitating the formation of melting in the copper. Consequently, a larger molten zone was formed at the glass/copper interface, leading to a stronger bonding strength. This observation explains why the negative focal plane position exhibited the highest bond strength, while the positive position showed the lowest strength in Fig. 3. Unless stated otherwise, the subsequent discussions and results are based on the focal plane position of – 0.5 mm.

In Fig. 3, it is evident that a minimum pulse energy of approximately 0.5 mJ was necessary for successfully creating a weld. Below this threshold, the laser energy was insufficient for generating an effective weld pool. It is important to note that the threshold pulse energy was higher than that required for fs lasers and ps lasers. This was because shorter pulse durations have higher instantaneous energy intensity, resulting in quicker local melting, albeit with a smaller resulting weld zone [6, 17]. As pulse energy increased, bond strength also increased due to the enlargement of the weld zone. However, as previously mentioned, a saturation region existed in the relationship between bond strength and pulse energy. After reaching a certain maximum, the strength remained unchanged and did not increase with further increases in pulse energy.

Possible explanations for this phenomenon are as follows: along with enlarging the molten zone, the melt's temperature also increased with increasing pulse energy. If there were hot spots in the melt with local temperatures higher than vaporization or even ionization temperature, voids or cracks could be generated in the resulting weld zone, resulting in residual stress and cracks along the weld interface and/or inside the weld. Although the weld zone could still be enlarged, the overall weld strength was not increased simultaneously. Additionally, the laser absorption coefficient of glass increases with temperature, especially for molten glass, indicating that less incident laser energy could reach the copper substrate. More absorbed energy in the glass facilitated the formation of non-uniform hot spots in the melt and decreased the final weld strength. Thus, it was critical to achieving a well-regulated thermodynamic status of the melt by properly partitioning incident laser energy using the focal plane position to achieve optimal weld strength in ns laser transmission glass/metal welding.

Furthermore, as depicted in Fig. 3, it was observed that the weld strength could reach up to 10 MPa, an achievement comparable to the strength values produced by an fs laser [13]. Consequently, by optimizing the process parameters, one can attain the required strength for welding glass to copper through ns laser transmission welding—an economically advantageous configuration.

3.2 Weld seam morphology and analysis

We first observed the width, W, and height, H, of the weld seams through cross-sectional CLSM images. To minimize end effects, a transverse cut was made in the middle portion of the weld line. The four images illustrated in Fig. 4 were obtained using the following four laser processing parameter sets. The laser powers were $P = 12$, 13, 14, and 15 W for Fig. 4 (a), 4 (b), 4 (c), and 4 (d), respectively. The scanning speed and pulse repetition rate were fixed at $v = 15$ mm/s and $f = 20$ kHz, respectively. The resulting processing pulse overlapping factor was 97.92%. The increase in power implied an increase in the incident energy per unit area (fluence, $F = \frac{P}{vD}$) and the corresponding fluences for the four powers were 20.00, 21.67, 23.33, and 25.00, respectively.

Figure 4 demonstrates that the size of the weld increased as the laser power was raised. When the laser pulse energy was increased from 0.6 to 0.75 mJ, the range of seam dimensions widened, with the width ranging from 50 to 72 µm and the height ranging from 46 to 59 µm. In contrast, achieving a similar increase in seam dimensions in the case of glass-to-aluminum transmission welding using a millisecond pulsed laser required a laser energy enhancement of approximately 200 mJ [20]. This disparity

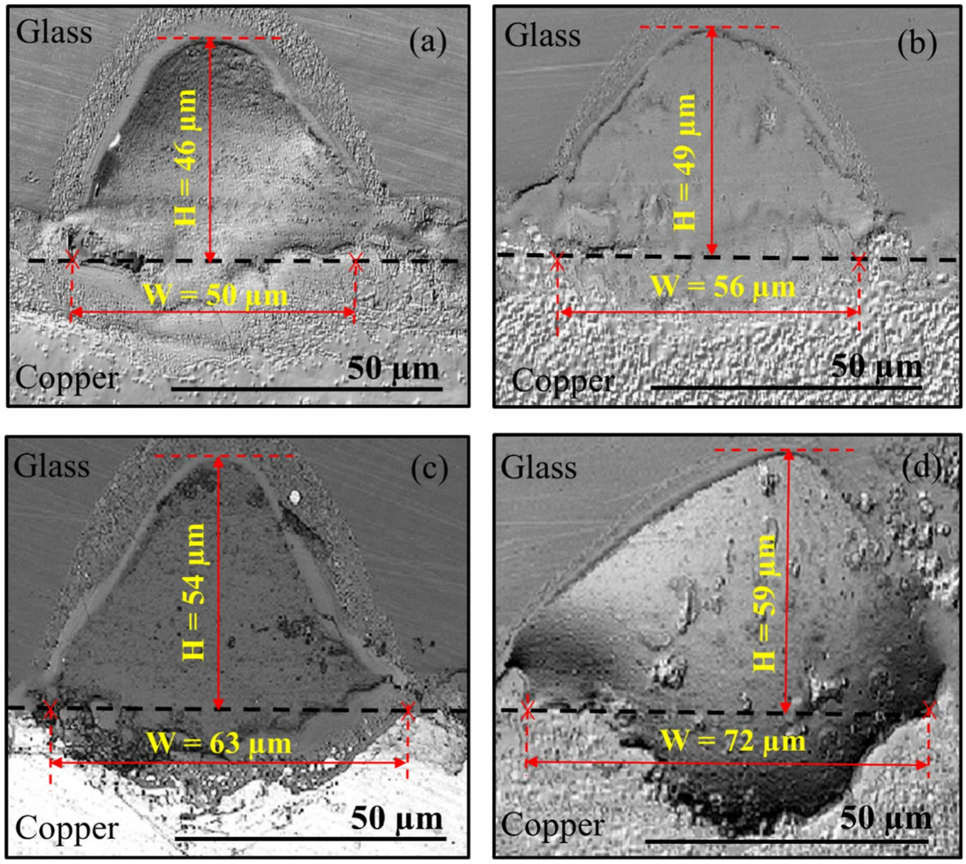

Fig. 4 Cross-sectional surface images of welds fabricated with various laser processing parameters were captured using confocal laser scanning microscopy. These images depict the internal structure of the welds before separation, which was carried out using a shear tensile separation force of 7.5 N. **a** $P = 12$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.6$ mJ; **b** $P = 13$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.65$ mJ; **c** $P = 14$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.7$ mJ; and, **d** $P = 15$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.75$ mJ

can be attributed to the higher intensity and shorter duration of action of the ns laser compared to those of the millisecond pulsed laser.

The cross-sectional images revealed that the majority of the weld zone is composed of glass, a phenomenon attributable to two key factors. Firstly, the exceptional transparency of glass to the incident laser light plays a pivotal role. As the laser irradiated the target, a substantial portion of the incident energy was absorbed by the copper surrounding the glass/copper interface, rapidly elevating the local copper temperature. Subsequently, this energy was conducted into the glass and the adjacent copper sections. Given that copper has a higher melting point of 1083 °C compared to the glass's softening point of 971 °C, the localized temperature surpassing the glass's softening point led to the initial deformation of the glass near the interface. The continued laser energy influx raised the temperature further, expanding the softened region within the glass. When temperatures exceeded the melting point of copper, it resulted in the melting of the interfacial copper. The second factor contributing to the extension of the softened glass zone was the laser's incidence from the glass side, coupled with the nonlinear optical absorptivity of the high-temperature softened glass [12]. As the glass heated, it progressively became more opaque to the incident laser light. This led to a greater absorption of laser energy by the glass, further enlarging the area of the softened and even molten glass zone.

One interesting observation is the location of the separation fracture when STSF is applied to separate the weld. Figures 5 (a) to 5 (d) show the top-view OM images of the line weld seams presented in Figs. 4 (a) to 4 (d), obtained after applying an STSF of 7.5 N along the weld seam's line direction. The top and bottom panels in each subfigure correspond to the images of the separated weld seams on the glass and copper sides, respectively. Three locations were examined on each side, and it was observed that the profiles on the glass side were convex, while those on the copper side were concave, indicating that a groove track appeared on the copper substrate and the main body of the weld resided on the glass substrates. Thus, the separation fracture occurred at the copper/weld interface.

On the other hand, when a PTSF of 7.5 N was applied perpendicular to the seam line direction for the weld separation, the corresponding top-view CLSM images, shown in Fig. 6 (a), (b), (c) and (d), demonstrated the opposite result, with the groove track appearing on the glass side, while the main body of the weld remained on the copper side after separation. Therefore, in this case, the separation fracture happened at the glass/weld boundary. The reasons for this difference are discussed below.

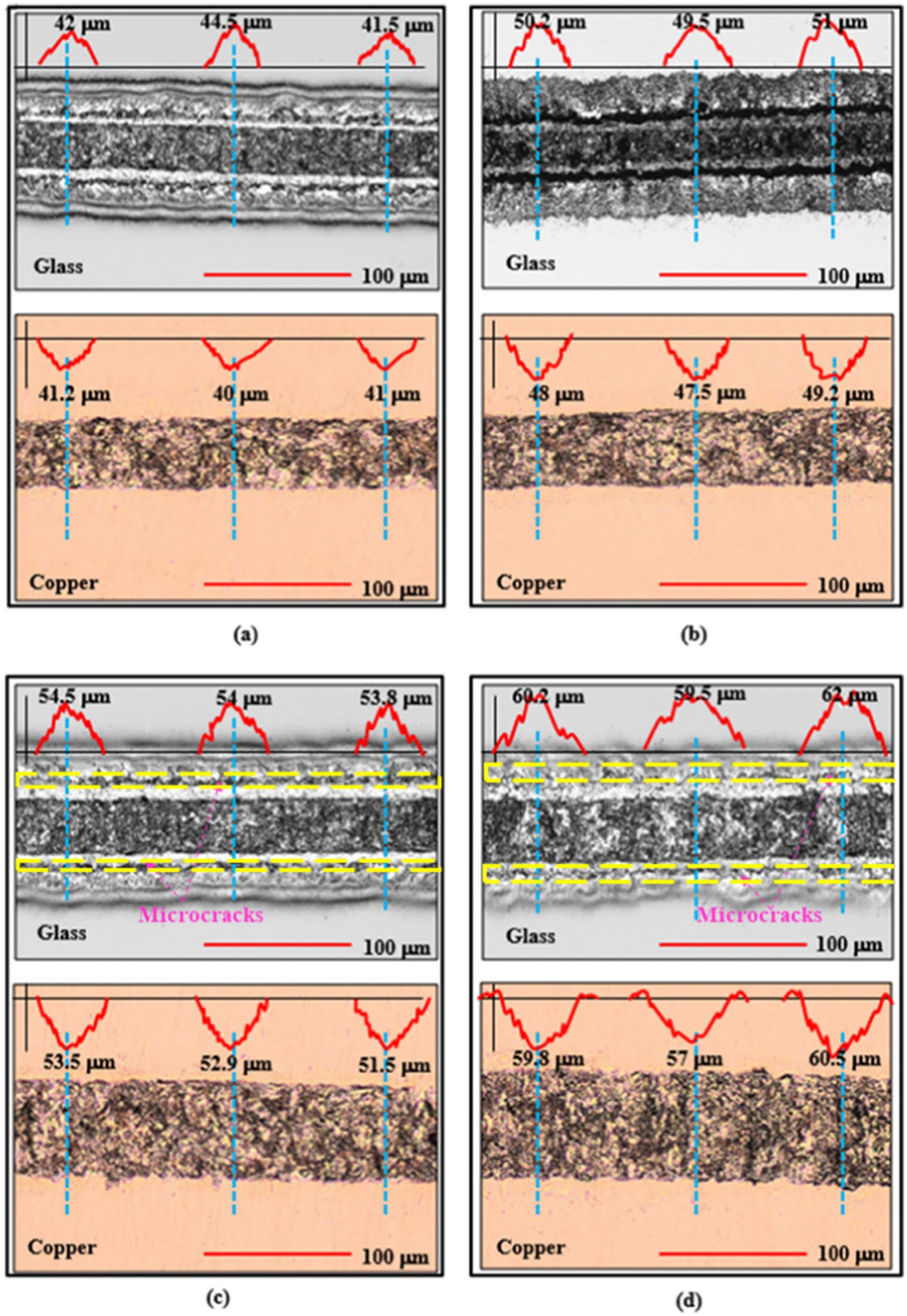

Fig. 5 Surface images of glass sheets and copper plates captured using a confocal laser scanning microscope after the separation of weld lines with a shear-tensile separation force of 7.5 N. The various fabrication laser processing parameters were: **a** P = 12 W, v = 15 mm/s, f = 20 kHz, and E = 0.6 mJ; **b** P = 13 W, v = 15 mm/s, f = 20 kHz, and E = 0.65 mJ; **c** P = 14 W, v = 15 mm/s, f = 20 kHz, and E = 0.7 mJ; and (**d**) P = 15 W, v = 15 mm/s, f = 20 kHz, and E = 0.75 mJ. The line profiles (indicated by blue dashed lines) scanned transversely across the weld line zone reveal that the weld remained on the glass sheet after separation, resulting in a convex profile on the glass sheet and a concave groove on the copper plate. The area containing microcracks is delineated by the yellow dashed line

Glass and copper are two fundamentally different materials, with glass being brittle and copper being ductile. Brittle materials can withstand high compressive stress but are much less tolerant to tensile stress, especially if there are defects or microcracks present. In contrast, ductile materials have a high tolerance to tensile stress. When the weld seam was subjected to a pull-tensile force, exerting along the z-axis and in the direction perpendicular to the weld line, both the weld/glass and the weld/copper interfaces shared the same projection area for sustaining the PTSS force, with the intercept area on the x–y plane. However, their interfacial extents were very different, as seen in the CLSM images in Fig. 4, where the perimeter of the weld/glass was much larger than that of the weld/copper. This larger contact area with the brittle material made the separation fracture more prone to occur around the glass side.

Additionally, the duration of the molten zone existence on the glass side was also much longer than on the copper

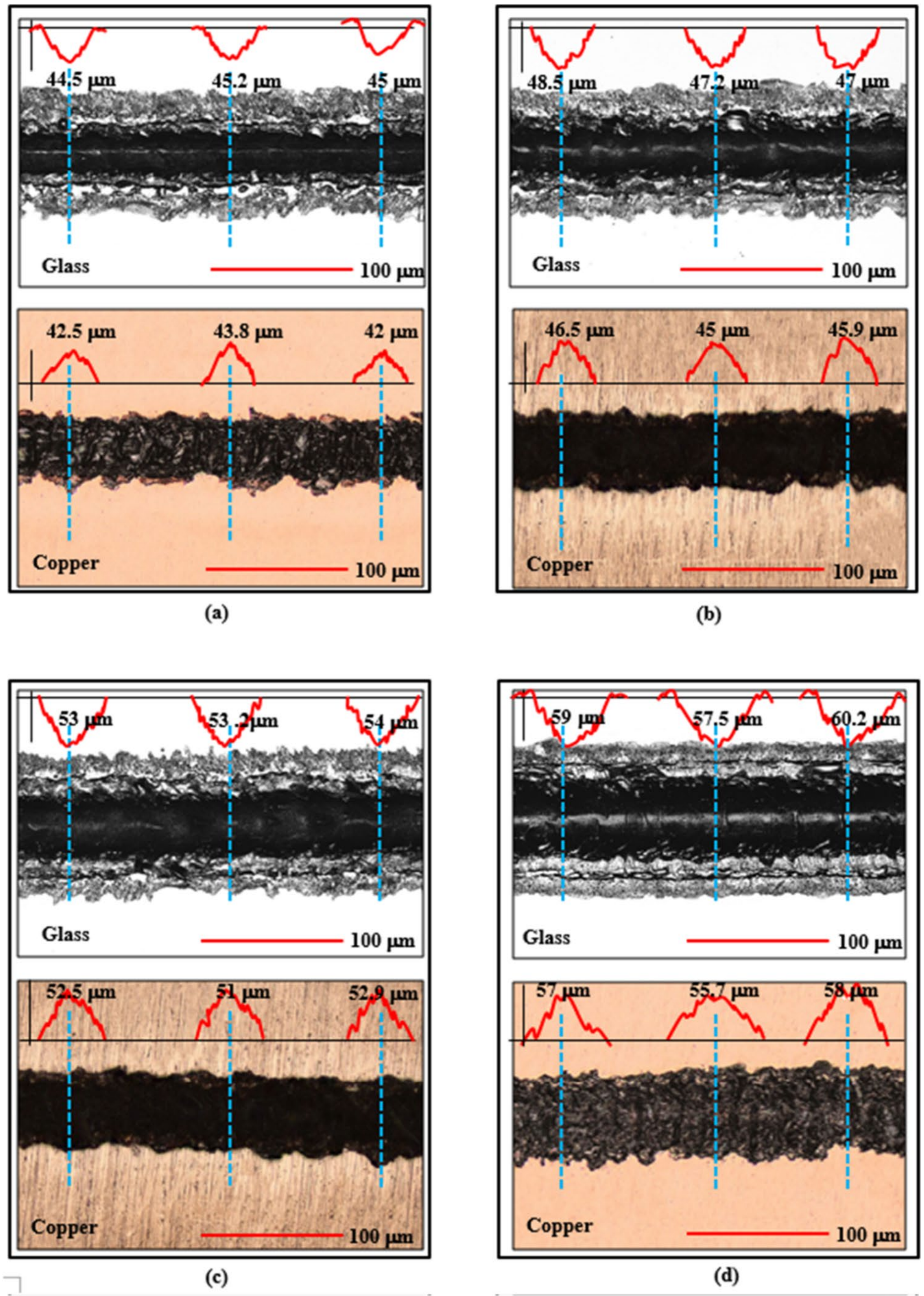

Fig. 6 Surface images of glass sheets and copper plates captured using a confocal laser scanning microscope after the separation of weld lines with a pull-tensile separation force of 7.5 N. The various fabrication laser processing parameters were: **a** $P = 12$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.6$ mJ; **b** $P = 13$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.65$ mJ; **c** $P = 14$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.7$ mJ; and (**d**) $P = 15$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E = 0.75$ mJ. The line profiles (indicated by blue dashed lines) scanned transversely across the weld line zone reveal that the weld remained on the copper plate after separation, resulting in a convex profile on the copper plate and a concave groove on the glass sheet

side due to the glass's lower melting point. As a result, if any microcracks were induced due to uneven changes in composition or temperature gradient during the melt resolidification, they would form around the weld/glass interface instead of the weld/copper interface. In Fig. 5, more observable microcracks, marked by the yellow dashed line, were present along the weld interface on the glass side. The presence of microcracks at the interface further reduced the material's strength and reduced its ability to withstand the PTSF.

On the other hand, when a STSF was applied to split up the two weld samples, the driving force for the separation of the weld from the matrix materials was the shear stress developed at the weld interface. As per the aforementioned discussion, the interfacial area of the weld/glass was much larger than that of the weld/copper, resulting in a smaller separation shear stress at the weld/glass interface compared to the weld/copper interface. Therefore, the separation fracture occurred on the copper side. As demonstrated in Fig. 6, where the groove tracks were on the copper side while the weld seams remained on the glass sides.

3.3 Elemental analysis and phase identification

The microstructures present in the weld and at the interface of the weld/matrix materials played a crucial role in determining the overall welding quality. To investigate these microstructures, a comprehensive analysis was conducted using various techniques, including SEM images, EDS analysis, TEM images, and electron diffraction patterns of the weld region. These techniques provided valuable insights into the composition, morphology, and crystal structure of the weld, enabling a thorough characterization of the microstructures involved in the welding process.

As depicted in Fig. 7(a) and its inset, a 10 μm × 10 μm area containing the glass/copper interface within the weld zone was precisely sliced using the FIB (focused ion beam) technique for microscale examinations and analysis. The white and black regions in the inset correspond to the copper and glass areas, respectively, reflecting the variations in the diffraction of emitted electrons due to the difference in atomic mass between copper and glass. The welding process was performed under the following set of process parameters: $P = 14$ W, $v = 15$ mm/s, $f = 20$ kHz, and $E_p = 0.7$ mJ.

Figure 7 (c), (d), and (e) illustrate the EDS mappings of the Cu, Si, and O elements, respectively, in the characterized area indicated in Fig. 7 (b). This area corresponded to a magnified view of the boxed region in the inset of Fig. 7 (a). The mappings revealed that the structure around the interface primarily consisted of blocky mixed regions of copper and glass. This indicated that convectional intra-mixing between copper and glass was the dominant mechanism for mixing within the molten pool during the welding process. The driving forces for this convection were mainly attributed to non-uniformly localized high pressure, which was induced by the rapid increase in temperature resulting from the absorbed incident laser energy. Additionally, spatial temperature and concentration gradients also contribute to the convectional mixing process. Furthermore, Fig. 7 (d) and (e) demonstrate that the distributions of Si and O elements exhibited a high degree of spatial consistency. This observation suggested that, in the majority of the glass present in the weld region, the atomic structure remained intact and was not disrupted during the welding process.

In addition to the intra-convention mixing, the presence of a more localized mixing mechanism, intra-diffusion, was also investigated by the examination of TEM images and SAED Confocal Laser Scanning Microscopy. Figure 7 (f), (g), and (h) display the SAED patterns of the circled regions A, B, and C, respectively, as indicated in Fig. 7 (b). These images were captured using an aperture of 150 nm. Region A, situated on the glass side and slightly further from the interface, exhibited a strong diffusion pattern in Fig. 7 (f), indicating the presence of only amorphous SiO_2 and no detectable copper nanoparticles in this region. Moving on to Region B, which was situated very near the center of the interface, Fig. 7 (g) exhibits various spot patterns and diffused rings, suggesting the presence of nanoparticles of copper nanoparticles, copper compound nanoparticles, and amorphous SiO_2 structures within this region. By utilizing the CrysTbox server 1.10 [29], the spot patterns surrounding the two distinct rings were identified as contributions from the polycrystalline structures of Cu_2O (011) and Cu (111) planes. The presence of Cu_2O (011) was further confirmed by calculating a d-spacing of 0.301 nm, which matched the corresponding value of Cu_2O in JCPDS No. 65–3288, and the occurrence of Cu (111) was verified by computing a d-spacing of 0.2087 nm, corresponding to JCPDS No.04–0836.

In comparison to Region B, Fig. 7 (h) reveals that the scattered spots in Region C were larger, indicating the presence of larger, single crystalline copper nanoparticles. Specifically, when considering the typical zone axis (011), two distinct spots were observed corresponding to Cu (111) and Cu (200) planes. The calculated d-spacing values for these spots, determined using ImageJ software, were 0.206 nm and 0.181 nm, respectively. These values closely matched the d-spacing values of 0.2088 nm for Cu (111) and 0.1808 nm for Cu (200) obtained from the reference pattern JCPDS No. 04–0836 (Cu) using the X'pert high-score plus 4.0 software.

Region B was selected for further nanostructure inspection in the weld. Figure 8 (a) presents the HR-TEM image of Region B, where two distinct interfacial locations, labeled 1 and 2, were analyzed. In Fig. 8 (b) and (c), the contrast between dark and light regions is evident, representing single crystal Cu nanoparticles and amorphous SiO_2, respectively. Notably, Fig. 8 (b) shows a gray lattice fringe near the SiO_2 boundary with a d-spacing of 0.21 nm, corresponding to the Cu_2O (200) plane. This observation suggested the formation of oxidized copper (Cu_2O) resulting from the reaction between copper and oxygen sourced from SiO_2 during the intra-diffusion process between Cu and SiO_2. Conversely, Fig. 8 (c) reveals the presence of an amorphous layer, several nanometers in size, between the Cu nanoparticles and amorphous SiO_2, despite the absence of a copper oxide layer. Drawing from the insights of a previous study [30] on the formation mechanism of a new phase, it could be concluded that this region consisted of a Cu–O-Si composition. The formation of the layer between Cu nanoparticles and amorphous SiO_2 was a significant contributing factor to the enhanced weld strength between Cu and SiO_2.

In this study, XRD measurements were employed to identify the potential phases present in the weld zone. Five different central positions on the cross-sectional plane of the weld zone, transverse to the line weld seam, were

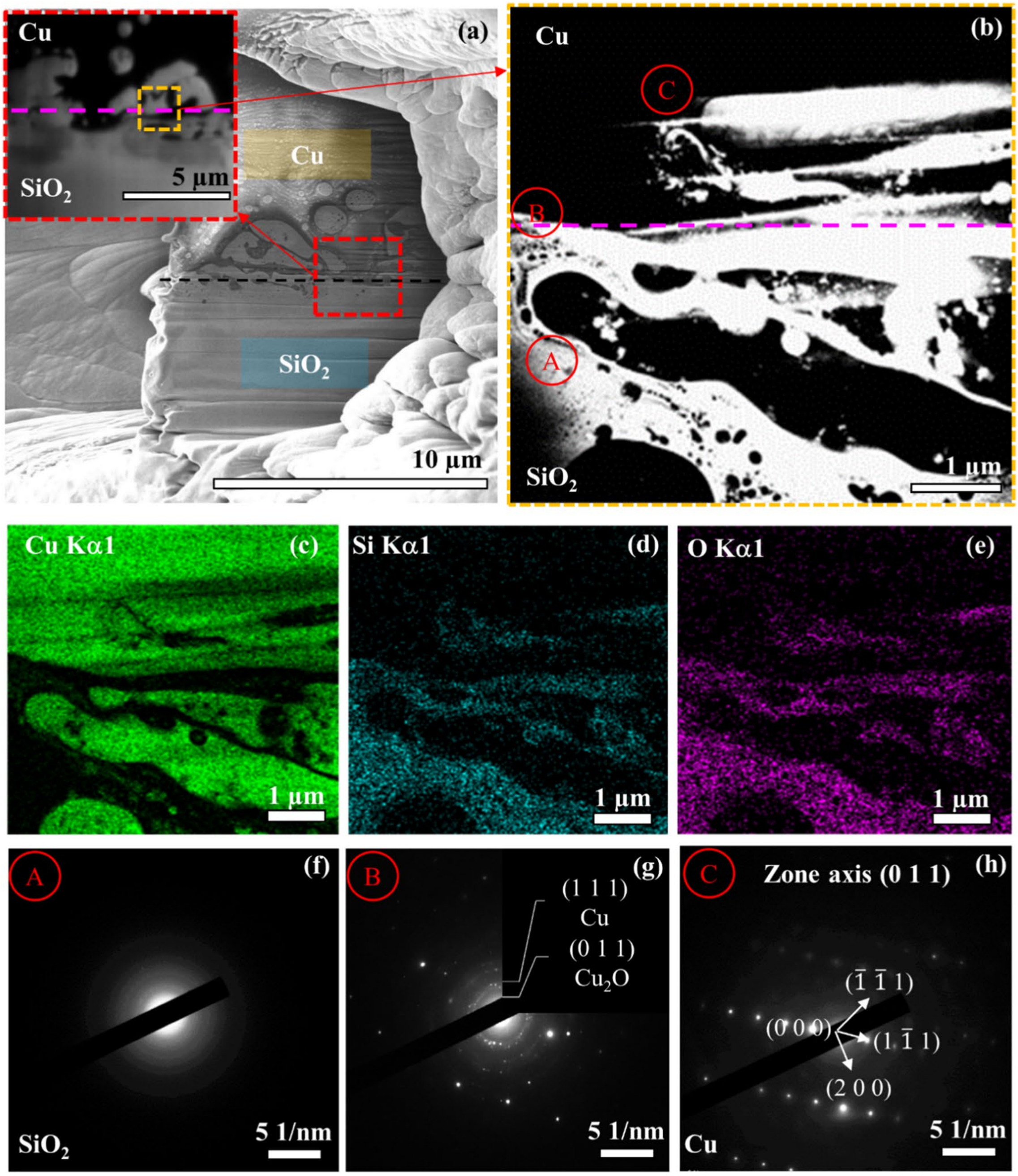

Fig. 7 Characterization of the microstructure, elemental compositions, and phase configurations around the interface of the Cu/SiO_2 weld using SEM, TEM, EDS, and SAED. **a** SEM image illustrating the FIB cut zone and the specific area (enclosed by the orange dashed line in the inset) selected for subsequent TEM, EDS, and SAED analyses. **b** Magnified SEM image of the enclosed zone from (**a**). Three circled zones, labeled as A, B, and C, were chosen for SAED observation. **c**, **d**, and **e** represent the EDS mappings of Cu, Si, and O elements, respectively, corresponding to (**b**). SAED patterns for the three circled regions A, B, and C in (**b**) are depicted in (**f**), (**g**), and (**h**), respectively

measured. The XRD spot size used was approximately 0.6 mm, and the sample was prepared using the same set of process parameters as shown in Fig. 7. The average XRD pattern obtained is depicted in Fig. 8 (d) and indicates the presence of Cu, CuO, and Cu_2O phases. The dominant phase identified was Cu, with diffraction peaks corresponding to the (111), (200), and (220) planes. However, the presence of the tenorite phase (monoclinic CuO) and the cuprite phase (cubic Cu_2O) appeared less distinct due to their smaller size and lower volume fraction in the selected region.

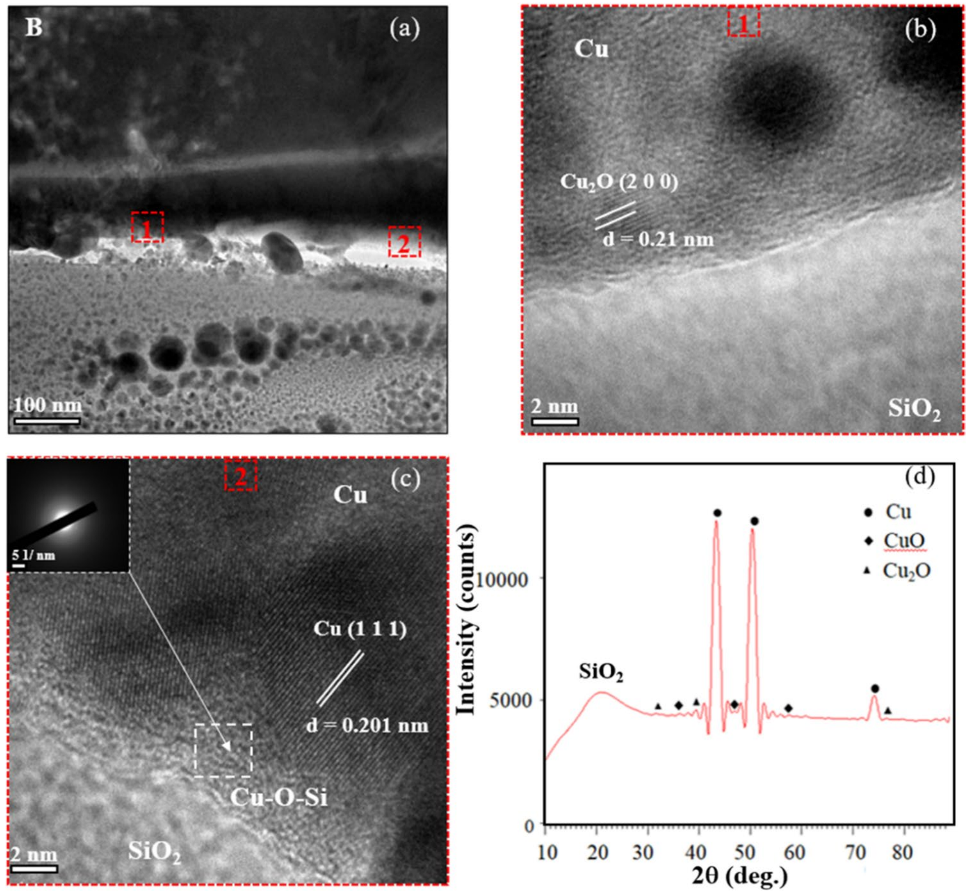

Fig. 8 Interfacial TEM images and XRD analysis. **a** TEM image focusing on the interfacial region around B, as indicated in Fig. 7 (b). **b** HR-TEM image of Area 1, enclosed by a red-dashed box labeled 1 in (**a**), for observing interfacial phase configurations. **c** HR-TEM image of Area 2, enclosed by the red-dashed box labeled 2 in (**a**), for observing interfacial phase configurations. **d** XRD patterns of the spotted Cu/SiO_2 weld interface

4 Conclusions

This study aimed to investigate the bonding process between a C1100 copper plate and an alkali-free glass sheet using a ns laser in laser transmission welding. The processing parameters, including laser pulse energy, repetition rate, and scanning speed, were systematically adjusted. The effects of three focal plane positions, namely above, at, and below the glass-copper interface, on the resulting bonding strength were examined. The weld strength was evaluated using two types of separation forces: pull-tensile force (PTF) and shear-tensile force (STF). The different fracture/separation results and the mechanisms causing them were observed, analyzed, and discussed. To understand the weld formation mechanisms and their correlation with the morphology, microstructure, elemental composition, and phase configuration around the weld interfaces and within the weld itself, various characterization techniques were employed. These included Confocal Laser Scanning Microscopy (CLSM), scanning electron microscopy (SEM), energy-dispersive spectroscopy (EDS), high-resolution transmission electron microscopy (HR-TEM), and selected area electron diffraction (SAED).

Based on the findings, the following major results were obtained and summarized.

1. With an increase in laser welding energy, the weld strength exhibited an initial improvement until reaching a peak, beyond which it remained constant. This saturation phenomenon can be attributed to the formation of voids or cracks, leading to the accumulation of residual stress and cracks within the weld zone. The saturation stage was influenced by the augmented presence of non-uniform hot spots in the molten area.
2. Positioning the focal plane below the glass/copper interface and on the copper side led to higher weld strength compared to positioning above or at the interface. This was because it ensured greater absorption of the laser energy in this setting. In laser transmission welding, the copper material primarily absorbed the laser energy, which then conducted through the glass, creating a molten zone and enabling successful welding.
3. The maximum bonding strength achieved was approximately 10 MPa, demonstrating ample strength for various applications necessitating the bonding of copper to glass. Remarkably, the weld exhibited superior resistance to shear-tensile separation force (STSF) when com-

pared to pull-tensile separation force (PTSF). During PTSF testing, the separation occurred at the juncture between the weld seam and the glass, leaving the weld seam intact on the copper plate. Conversely, under STSF conditions, separation occurred along the weld-to-copper interface, leaving the weld seam adhered to the glass sheet.

4. The cross-sectional image of the CLSM revealed that the weld seam was predominantly occupied by glass. This can be attributed to the lower melting point of glass compared to copper, leading to a larger molten zone of glass. Furthermore, at higher temperatures, glass demonstrates an increased light absorption capacity, thereby magnifying the size of its molten zone. The expanded interface between the weld and glass increased the likelihood of fracture occurring along this boundary under PTSF.
5. The analysis of SEM images and EDS mappings provided evidence that intra-mixing and inter-particle diffusion of Cu and SiO_2 played vital roles during the bonding process. Notably, the HR-TEM and SAED observations revealed the existence of polycrystalline copper nanoparticles, Cu_2O, and an amorphous Cu–O–Si region at the weld interface. This might be critical in enhancing the quality of welding between copper and glass.

Authors' contributions Hoai Nguyen: Investigation, Experiment, Data curation, Writing-original draft, Visualization.

Chih-Kuang Lin: Conceptualization, Methodology, Discussion, Modification.

Pi-Cheng Tung: Conceptualization, Methodology, Discussion, Modification.

Jeng-Rong Ho: Resource, Conceptualization, Methodology, Writing-review & editing, Supervision, Funding acquisition.

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Data availability The data and material is available.

Code availability 'Not applicable' for this section.

Declarations

Ethics approval This ethics are approved.

Consent to participate 'Not applicable' for this section.

Consent for publication We consent for publication.

Conflicts of interest/Competing interests There are no Conflicts of interest/Competing interests.

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